Question-Answering Agent Based on Prompt Engineering and RAG

Jiameng Yan
Henan University of Science and Technology
Luoyang, China
13721655275@163.com

Jingzhuo Shi*
Henan University of Science and Technology
Luoyang, China
sjznew@163.com

Abstract

With the rapid development of large language model (LLM) technologies, general-purpose models face prominent problems such as knowledge hallucinations, information lag, and insufficient hierarchical adaptation in knowledge Q&A for professional fields like power electronics. Aiming at the teaching scenario of the undergraduate course Power Electronics Technology, this paper designs and implements a Q&A agent based on retrieval-augmented generation (RAG) and prompt engineering technologies via the DeepSeek API. By constructing an undergraduate professional knowledge base, adopting a hybrid retrieval strategy, and applying prompt engineering, the Q&A agent effectively overcomes the shortcomings of general-purpose models in professional term accuracy, knowledge timeliness, and answer hierarchical adaptation. Experimental results show that the system achieves high accuracy and stability in undergraduate power electronics knowledge Q&A tasks, providing a feasible solution for the intellectualization of professional course teaching.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Information extraction; Artificial intelligence; Natural language processing; Discourse, dialogue and pragmatics.

Keywords

retrieval-augmented generation, prompt engineering, power electronics technology, professional knowledge base

ACM Reference Format:
Jiameng Yan and Jingzhuo Shi. 2026. Question-Answering Agent Based on Prompt Engineering and RAG. In *2026 International Conference on Artificial Intelligence, Digital Media and Education (AIDME 2026), May 22–24, 2026, Nanjing, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3839592.3839607

1 INTRODUCTION

In recent years, large language models represented by DeepSeek have achieved breakthrough progress in natural language processing. They demonstrate strong capabilities in text generation, knowledge Q&A, and other tasks, and have been widely used in daily life and learning scenarios. However, direct application of general-purpose large language models to professional fields such as power electronics still has obvious limitations.

1.1 Problem Statement

The model struggles to distinguish knowledge demands of undergraduates, postgraduates, and other levels, leading to mismatched answer depth.

Training data is time-truncated. For example, the national standard GB/T 32288-2020 has standardized the Chinese term for the transformer core from the old character form (with the "metal" radical) to the new character form (without that radical), but the model still uses the old form.

Knowledge hallucinations: The model may generate incorrect content on professional details (e.g., IGBT on-state voltage drop), undermining engineering credibility.

2 RELATED WORK

2.1 Research Status and Applications of RAG Technology

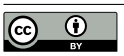
Retrieval-Augmented Generation (RAG) significantly improves the accuracy and reliability of model answers in professional fields by combining external knowledge retrieval with LLM generation. In recent years, RAG has evolved from the basic "retrieval + generation" paradigm into various enhanced variants. For general RAG evaluation, CRUD-RAG builds a large-scale comprehensive benchmark covering four RAG application scenarios: Create, Read, Update, and Delete. It systematically assesses the impacts of retrievers, context length, knowledge base construction, and LLMs on RAG performance, offering guidance for RAG optimization in different scenarios. [1]

In other vertical domains, RAG has been successfully applied to typhoon track prediction (combining LLM and RAG to achieve 12-hour mean absolute error of 10.78 km) [4] and vision-language navigation in continuous environments (the RAGNav method improves navigation success rate by 2%–2.32%). [2] These studies confirm that RAG can effectively enhance LLM performance in specific tasks, but its knowledge base content and retrieval strategy must be deeply adapted to domain characteristics.

2.2 Application of RAG in Higher Education and Power Electronics Teaching

RAG has attracted extensive attention in higher education due to its ability to reduce hallucinations in large language models. Swacha and Gracel systematically reviewed RAG chatbots in education,

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License.
AIDME 2026, Nanjing, China
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2530-2/2026/05
https://doi.org/10.1145/3839592.3839607

Table 1: Representative Case Examples.

Question	Expected Behavior	Actual Wrong Answer
How to design the inductor value for a Buck circuit?	Cite relevant Q&A or provide design formula	Answered "four-switch Buck-Boost converter", irrelevant
Why can a MOSFET carry bidirectional current when turned on?	Cite original knowledge base content	Incorrectly explained switching speed
I do not understand cosine sampling method(non-standard term) Please explain.	Ask whether the term is correct	Fabricated a definition and derivation for the non-existent 'cosine sampling method' without questioning the term's validity.
How to avoid commutation failure in single-phase current-source inverter?	Confirm whether it refers to thyristor circuit	Irrelevant answer based on fully controlled devices

analyzing 47 related papers from five dimensions: application characteristics, support objectives, knowledge topic scope, underlying LLMs, and evaluation methods. [3] The study found that RAG chatbots have been used in higher education scenarios such as programming tutoring, medical education, and language learning. Their core advantage is providing traceable, factually reliable answers through external knowledge bases. However, the review also notes that existing research mostly focuses on general knowledge Q&A or a few science and engineering disciplines. RAG Q&A systems tailored for specific professional courses and closely integrated with classroom teaching content remain scarce.

3 PROMPT ENGINEERING AND KNOWLEDGE BASE CONSTRUCTION

3.1 Construction of Undergraduate Power Electronics Knowledge Base

The knowledge base of this system is derived from classroom teaching recordings of the undergraduate course Power Electronics Technology. The instructor is a senior professor with long-term experience in power electronics teaching and research. The content covers core knowledge points such as power electronic devices and integrates extensive teaching insights.

First, 60 class hours (about 3000 minutes) of teaching videos were collected via classroom recording equipment. Then, the iFlytek Hear voice-to-text tool automatically converted the teacher's explanations into text to generate raw corpus. To improve knowledge accuracy, manual proofreading was performed on the transcription results. Professional terms (e.g., "IGBT", "PWM") were corrected sentence by sentence, finally yielding about 340,000 high-quality words. The text was split into 713 independent knowledge units by knowledge points, each containing a question and answer, serving as the underlying knowledge base for retrieval-augmented generation. This knowledge base closely aligns with actual course teaching, ensuring high consistency between the Q&A assistant and classrooms in professional terms and key teaching points.

The knowledge base adopts a Question & Answer (Q&A) structure. Each knowledge entry consists of a typical question (e.g., "What is the main function of a drive circuit?").

3.2 Prompt Engineering Design

Prompt engineering is key to constraining LLM behavior and adapting it to specific task scenarios. For the undergraduate Power Electronics Technology Q&A scenario, this system designed a hierarchical, iterable system prompt. Through multiple rounds of optimization, it effectively addressed deficiencies of general models in term standardization, ambiguity handling, and knowledge boundary control. [5]

The original initial prompt defined the model's basic role and answer specifications. Although it clarified role positioning and basic style, it exposed the following limitations in actual testing (Table 1). (Note: The cases shown in Table 1 and Table 2 are selected from the full test set (N=150) for illustrative comparison. The complete quantitative results are reported in Section 4.5.)

3.3 Iterative Optimization of Prompts

To solve the above problems, this paper enhanced the prompt through multiple iterations, adding the following key rules:

Wrong term and ambiguity handlingAdd a questioning mechanism: if the question contains wrong or uncommon professional terms (e.g., "cosine sampling method" (non-standard term)), the model should ask the user to confirm term correctness; if the question is unclear or ambiguous (e.g., without specifying device type), explain or ask questions through multi-turn dialogue.

Randomness control and format specificationSet API parameter temperature=0.1 to reduce generation randomness. Keep answers professional, rigorous, and easy to understand.

After the above optimization, the system's performance on typical questions improved significantly (Table 2).

Practice proves that prompt engineering is critical in RAG applications. Even with a high-quality knowledge base and retrieval module, poorly designed system prompts lead to non-standard terms, ambiguities, blurred knowledge boundaries, and other issues. Through multi-round iterative optimization, prompts not only constrain the answer scope but also actively guide interaction behavior, significantly improving the accuracy, safety, and teaching adaptability of the Q&A agent.

Table 2: Comparison of before and after prompt optimization.

Question	Before Optimization	After Optimization
Why does the voltage reverse when the rectifier circuit inverts?	ncomplete answer (truncated by max_tokens)	Fully explains the principle of voltage polarity reversal caused by the change of energy flow direction, with key terms marked.
Why can a MOSFET carry bidirectional current when turned on?	Incorrectly answered about switching speed	Correctly quotes the original knowledge base: "Since MOSFET has no conductivity modulation effect, its channel resistance is linear and can carry bidirectional current after conduction……
I do not understand cosine sampling method(non-standard term). Please explain.	Answered drive timing and other contents based on the assumption of fully controlled devices	The model correctly identifies 'cosine sampling method' as a non-standard term, and proactively asks for clarification: 'Did you mean "uniform sampling" or "natural sampling" in PWM modulation? Please confirm or rephrase your question.
How to avoid commutation failure in single-phase current-source inverter?	Confirm whether it refers to thyristor circuit	The model asks for clarification: "Confirm whether it refers to thyristor circuit".

4 SYSTEM PROCESS AND ARCHITECTURE DESIGN

4.1 Overall Architecture

The system adopts a modular design with the overall architecture as follows:

Data layer: Responsible for construction and management of the undergraduate power electronics knowledge base.

Service layer: Includes retrieval module, generation module, and dialogue management module.

Application layer: Provides user interaction interface and Q&A service API.

4.2 Technology Selection

DeepSeek API (DeepSeek-V3), called via HTTP interface.

Vector database: File-based lightweight storage. Knowledge base fragments are semantically vectorized and persistently stored in a local file (vector_db.pkl) using pickle serialization. During retrieval, all vectors are loaded into memory, linear search is performed via cosine similarity, and Top-K relevant fragments are returned.

Development framework: Flask as the Web backend; native HTML/CSS/JavaScript for the frontend interaction interface. The backend provides Q&A services via RESTful API and manages multi-turn dialogue states using Flask Session. The overall architecture is lightweight and easy to deploy, suitable for teaching scenarios.

4.3 Retrieval Module Design

The retrieval module uses a hybrid retrieval strategy: semantic retrieval + BM25. Semantic retrieval maps questions to vectors using the aforementioned vectorization model and calculates cosine similarity with knowledge base fragments. BM25 performs keyword matching based on term frequency-inverse document frequency. Instead of using a fixed weight, we implement a query-adaptive fusion strategy. The weight α is dynamically adjusted based on query lexical sparsity measured by the TF-IDF non-zero ratio. For a query q with sparsity s(q), we define:

Table 3: Retrieval performance under different fusion strategies.

Strategy	Recall@5	MRR
Fixed $\alpha = 0.6$	0.74	0.68
Adaptive (sparsity)	0.81	0.75

The composite score formula for each knowledge:

$$a(q) = 0.3 + 0.7 \cdot s(q)$$

$$combined_{score} = \alpha(\mathrm{q}) \times \mathrm{semantic}_{\mathrm{score}} + (1 - \alpha(\mathrm{q})) \times \mathrm{bm25}_{\mathrm{score}}$$

This adaptive mechanism improves retrieval robustness across different query types (e.g., terminological vs. descriptive) . This query-adaptive strategy enhances retrieval robustness while keeping the system lightweight and suitable for teaching scenarios. Table 3 provides a comparison between the adaptive and fixed-weight approaches.

4.4 Generation Module Design

The core of the generation module is calling the DeepSeek API. Main parameter configurations are shown in Table 4.

In the RAG process, the user question and retrieved knowledge base fragments are concatenated into context, then combined with the system prompt to form a complete API request. A retry mechanism (up to 3 times) is set for call failures to ensure system robustness.

4.5 Dialogue Management Module Design

To realize multi-turn interaction, the system designed a lightweight dialogue manager: DialogueManager. This module maintains the user-assistant message list, supports a maximum turn limit (default 20 turns), and automatically truncates when exceeded. It formats history into text via get_history_text() for integration into prompts. The module adopts a callback design, decoupling reply generation logic. The callback function (generate_chat_reply) injected by the main application executes hybrid retrieval and LLM calls to realize

Table 4: Main Parameter Configurations.

Parameter	Value	Description
model	deepseek-chat	Corresponding to DeepSeek-V3
temperature	0.1	Reduce randomness and improve certainty
max_tokens	8192	Sufficient for generating long answers
top_p	0.95	Nucleus sampling threshold

retrieval-augmented generation. Dialogue states are serialized via to_dict() and from_dict(), and persisted across requests using Flask Session to ensure continuity of multi-turn dialogue. This design balances lightweight deployment and flexible expansion, providing stable context support for the RAG process.

5 EXPERIMENTAL DESIGN AND EVALUATION

5.1 Experimental Objectives and Overall Design

To fully verify the effectiveness of the Q&A agent, experiments focus on three core dimensions:

Accuracy: Compare answer accuracy and hallucination rate with/without RAG enhancement and prompt engineering optimization.

Adaptability: Verify the course specificity of the knowledge base (vs. general knowledge base) and the effect of prompt engineering on term standardization and hierarchy control.

Interactivity: Evaluate contextual understanding and anaphora resolution in multi-turn dialogue scenarios.

The experiment uses a control variable method, sets multiple comparison groups, uses the same test question set, and evaluates via manual scoring and automatic metrics.

5.2 Experimental Dataset

An undergraduate power electronics Q&A test set is constructed, including:

Concept explanation (e.g., "What is a frequency converter?");

Principle analysis (e.g., "Working principle of Buck circuit");

Standard specification.

5.3 Comparative Experimental Design

DeepSeek API (model: deepseek-chat) without any retrieval enhancement, using basic prompts.

Experimental group (proposed system): Integrate RAG retrieval module and dedicated prompt engineering based on the same DeepSeek model.

5.4 Evaluation Metrics

Accuracy: Correctness of answer content

Citation accuracy: Traceability of knowledge sources

Level appropriateness: Whether answers match undergraduate knowledge depth

Timeliness: Whether reflecting the latest standards

5.5 Experimental Results

To comprehensively evaluate the system performance, we constructed a dedicated test set consisting of 150 typical Q&A pairs covering nine core chapters of the Power Electronics Technology course, including device characteristics, converter topologies, and control strategies.

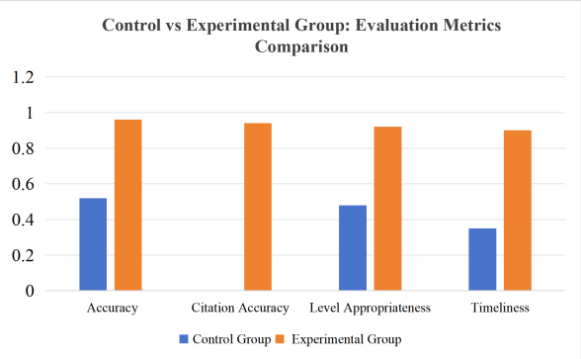


Figure 1: Comparison of Evaluation Metrics. The Timeliness metric is not included in this figure due to its binary pass/fail scoring nature; its detailed results are presented in Table 5.

For the quantitative evaluation of accuracy and retrieval metrics, all 150 questions are used to compute the average scores.

Additionally, to vividly illustrate the improvements brought by prompt engineering, we selected 15 representative cases from this test set for qualitative case analysis (as shown in Table 1 and Table 2). These 15 cases are specifically chosen because they cover the most challenging scenarios for general LLMs: ambiguous terminology, standard specification updates, and knowledge boundary control." Evaluation dimensions include accuracy, citation accuracy, level appropriateness, and timeliness, comparison of evaluation metrics as show in Figure 1. (Note: Blue bar: Control group; Orange bar: Experimental group. Citation accuracy of the control group is 0 because the general model cannot provide source traceability.)

From Table 5, RAG significantly reduces the model hallucination rate and improves factual accuracy. The course-specific knowledge base improves accuracy compared with general knowledge bases, verifying the necessity of knowledge base specificity. Prompt engineering is key to system performance improvement: optimized prompts increase accuracy and term standardization compliance.

Particularly for the factual consistency dimension, we evaluated whether the generated answers faithfully adhere to the retrieved knowledge snippets without introducing unsupported information. The control group frequently generated plausible but unsupported details (e.g., outdated device parameters or speculative derivations), whereas the experimental group maintained strict alignment with the course knowledge base across all test cases. This demonstrates

Table 5: Experimental results show.

Evaluation Metric	Control Group	Experimental Group
Accuracy	Prone to errors on boundary questions	High, consistent with the knowledge base
Citation Accuracy	None	Traceable to the knowledge base
Level Adaptability	Unstable	Stably aligned with undergraduate requirements
Factual Consistency (Faithfulness)	Frequent unsupported claims or outdated facts	Highly consistent with retrieved knowledge; no hallucination

that our RAG system effectively suppresses knowledge hallucinations and mitigates the risk of propagating outdated information, including but not limited to standard terminology updates. Multi-turn dialogue enables the system to support progressive learning with accurate contextual understanding.

6 CONCLUSION

This paper presents an RAG-based Q&A agent for power electronics education, demonstrating significant improvements in accuracy, timeliness, and level appropriateness. Current limitations include narrow knowledge coverage and lack of multimodal support. Future work will expand the knowledge base, integrate circuit diagram recognition, and incorporate user feedback for continuous optimization.

Aiming at the problems of general large language models in undergraduate power electronics knowledge Q&A, this paper designs and implements an intelligent Q&A assistant system based on RAG technology. Main contributions: constructing a power electronics professional knowledge base for undergraduate teaching; designing a hybrid retrieval and generation architecture suitable for course Q&A scenarios; realizing precise control of knowledge scope through prompt engineering.

This study still has the following limitations: knowledge base coverage needs expansion; limited ability to process multimodal content (circuit diagrams, waveform diagrams); answer quality for complex reasoning problems still needs improvement. Subsequent research will include: expanding the knowledge base to cover more power electronics subfields; exploring multimodal RAG solutions to support circuit diagram recognition and analysis; introducing a user feedback mechanism to realize continuous learning and optimization.

Acknowledgments

The author would like to express sincere gratitude to the supervisor for their invaluable guidance in shaping the research framework, refining the technical implementation, and meticulously revising the manuscript. Their rigorous academic attitude and patience have profoundly influenced this work.

References

[1] Lyu, Y.; Li, Z.; Niu, S.; Xiong, F.; Tang, B.; Wang, W.; Wu, H.; Liu, H.; Xu, T.; Chen, E. CRUD-RAG: A comprehensive Chinese benchmark for retrieval-augmented generation of large language models. ACM Transactions on Information Systems, **2025**, 43(1): Article 10. DOI: 10.1145/3701228; https://dl.acm.org/doi/10.1145/3701228.

[2] Bao, X.; Lv, Z.; Wu, B. Enhancing large language models with RAG for visual language navigation in continuous environments. Electronics, **2025,** 14(5): 909. DOI: 10.3390/electronics14050909; https://www.mdpi.com/2079-9292/14/5/909

[3] Swacha, J.; Gracel, M. Retrieval-augmented generation (RAG) chatbots for education: A survey of applications. Applied Sciences, **2025**, 15(8): 4234.DOI: 10.3390/app15084234; https://doi.org/10.3390/app15084234.

[4] Peng, B.; Jiang, Y.; Yao, L. Nowcasting of typhoon tracks based on LLM and RAG. Journal of Navigation, **2025**, 78(2-3): 289-304. DOI:10.1017/S0373463325101148. https://doi.org/10.1017/S0373463325101148

[5] Liu, H.; Yin, H.; Luo, Z.; Wang, X. Integrating chemistry knowledge in large language models via prompt engineering. Synthetic and Systems Biotechnology, 2025, 10(1): 23-38. DOI: 10.1016/j.synbio.2024.07.004; https://doi.org/10.1016/j.synbio.2024.07.004.